Moazzam Maan

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EDUCATION

B.S. in Computer Science, B.S. in Physics, B.S. in Mathematics
Roanoke College, Salem, VA — August 2023 — May 2027 (expected)
GPA: 4.00/4.00

RESEARCH EXPERIENCE

Finding Largest Independent Sets in Circulant Graphs — May 2024 — February 2026
Advisor: Dr. Anil Shende, Professor of Computer Science

- Used properties of independent sets in a circulant-graph model of a network of cellular towers to find an upper bound on the frequency separation between devices.
- Generalized these graphs into an infinite family and established bounds for the independence numbers of this broader family, while also determining the exact independence number in special cases.

Percolation Model of Spiral Galaxy Formation — December 2025 — Present
Advisor: Dr. Truong Le, Assistant Professor of Physics

- Recreated the Seiden–Schulman gas-regulated galaxy formation model in Python, to our knowledge the first open-source reproduction.
- Added observational rotation curve data to recreate spiral galaxy morphology.
- Showed that 1–10 initial star-forming clusters can evolve into large-scale galaxy structure, matching early-galaxy observations such as the Firefly Sparkle.

High–Altitude Balloon — January 2026 — Present
Advisor: Dr. Truong Le, Assistant Professor of Physics

- Worked on a nine-person interdisciplinary team to design, build, and launch a high-altitude balloon experiment.
- Wrote the electronics code for onboard data collection and reliable sensor functionality during flight.
- Assembled and integrated the flight electronics, including sensors, an Arduino microcontroller, and onboard video equipment.
- Supported a successful launch in which the balloon reached approximately 27 km and landed safely.
- Currently analyzing environmental and flight data collected during the launch.

Cops, Robber, and a Duck — December 2025 — Present
Advisor: Dr. Michael Weselcouch, Assistant Professor of Mathematics

- Introduced *Ducks and Robbers*, a pursuit–evasion graph theory game in which a duck acts as a dynamic blockade for both cops and robbers.
- Characterized 1-duck-cop-win graphs and determined duck-cop numbers for disconnected graphs, trees, and hypercubes.
- Constructed, for any pair k, p, a graph with cop number k and duck-cop number p.

3D Wildfire Visualization — February 2025 — September 2025
Advisor: Dr. Anil Shende, Professor of Computer Science

- Created a web-based software for visualizing 3D forest fire spread. Currently working on disseminating our product as open-source.
- Participated in the SERDP/ESTCP Virtual Wildfire Modeling and Immersive Training Prize Challenge and placed top 10 nationwide out of over 30 applicants.

Interactive C++ Learning Environment November 2025 — April 2026

Advisor: Dr. Durell Bouchard, Associate Professor of Computer Science

- Built a mini pedagogical C++ compiler and developed an interactive programming IDE in Minecraft.
- Designed programming lessons for CPSC-120 on sequential execution, loops, and conditional logic to support teaching in a game-based, interactive setting.

Statistical Analysis of Mario Kart 64 Race Data January 2025 — Present

Advisor: Dr. Michael Weselcouch, Assistant Professor of Mathematics

- Created a Lua pipeline to extract real-time in-game MK64 data through BizHawk emulator memory.
- Conducted a semester-long research project collecting gameplay data from student races.
- Currently developing a SCORE Network-style educational module using Mario Kart 64 race data for statistics and data science classes.

PUBLICATIONS

1. **Structural Analysis of Largest Independent Sets in a Family of Circulant Graphs** *(with A. Shende).*
   - Proceedings of Joint Mathematics Meetings (JMM) (2025).
   - Proceedings of the Consortium for Computing Sciences in Colleges (CCSC) (2024).
   - Complete manuscript in preparation.
2. **Web-browser-based Visualization of Wildland Fire Data** *(with Y. Satynska and A. Shende).*
   - Proceedings of the Consortium for Computing Sciences in Colleges (CCSC) (2025).
3. **Percolation Models of Spiral Galaxy Formation** *(with T. Le, and C. Amberman).*
   - Proceedings of the 104th Annual Virginia Academy of Science Meeting, *Virginia Journal of Science* (2026, forthcoming).

TALKS AND PRESENTATIONS

1. **Interactive Web-Based Visualization of Wildland Fire Data**
   - Virtual Wildfire Modeling and Immersive Training Prize Challenge, Orlando, Florida, September 2025 (presentation).
   - Undergraduate Research Showcase at the Capitol, Virginia General Assembly, February 2026 (poster).
   - Mathematics, Computer Science and Physics Conversation Series, Roanoke College, November 2025 (talk).
   - CCSC Student Research Showcase, Mercer University, November 2025 (poster).
2. **Finding Largest Independent Sets in Families of Circulant Graphs**
   - Joint Mathematics Meetings (JMM) Student Research Showcase, Seattle Convention Center, January 2025 (poster).
   - National Conference on Undergraduate Research (NCUR), Greater Richmond Convention Center, April 2026 (poster).
   - Mathematics, Computer Science and Physics Conversation Series, Roanoke College, October 2024 (talk).
   - CCSC Student Research Showcase, Furman University, November 2024 (poster).
3. **Percolation Model of Spiral Galaxy Formation**
   - Virginia Academy of Science Annual Meeting (VAS), Christopher Newport University, 2026 (poster).

HONORS AND AWARDS

- **CCSC Research Competition:** First Place (2024), Third Place (2025).
- **VAS Annual Meeting:** Best Poster Presentation (2026).
- **CCSC Programming Competition:** Third Place (2025).
- **Roanoke College Research Showcase:** Third Place (2024).
- **SERDP Virtual Wildfire Modeling Challenge:** Top 10 national placement (2025).
- **Roanoke College Summer Scholar:** 2025, 2026.
- **Honor Societies:** Pi Mu Epsilon; Upsilon Pi Epsilon.

PROFESSIONAL EXPERIENCE

Virtual Reality Development Intern February 2025 — July 2025
US Department of Agriculture - Forest Service Missoula, MT (Remote)

- Developed an immersive and interactive web-based tool for visualizing output from QUIC-Fire, a state-of-the-art wildfire modeling software developed at Los Alamos National Laboratory.
- Attended bi-weekly meetings of the 3D/XR Community of Practice to exchange ideas with fire visualization developers and firefighters across the country.

Teaching Assistant for Math and Computer Science August 2024 — December 2024
Computer Science Department, Math Department - Roanoke College Salem, VA

- Assisted in teaching first-year courses in classes and labs.
- Provided feedback to the professor about suggested problems for the students in the class.

Subject Tutor for Computer Science, Math and Physics Aug 2024 — Dec 2025
Center for Learning and Teaching - Roanoke College Salem, VA

- Earned a Level 1 Certification through the College Reading and Learning Association.
- Provided tutoring sessions for 10+ courses.

Resident Advisor and Community Leader August 2025 — Present
Residence Life and Housing - Roanoke College Salem, VA

- Supervise and maintain the safety of 25+ residential students.
- Organize, manage, and execute bi-weekly events available to up to 200 residential students.

Pass On Hunger Treasurer January 2024 — Present
Roanoke College Pass On Hunger - Food Distribution Salem, VA

- Use the value of students' leftover meal swipes to donate ingredients to the Roanoke Rescue Mission.
- Communicate with 400 students to collect $1,400 weekly donations and manage orders and drop-off.

SKILLS

- **Programming:** C++, Python, Java, Julia, Arduino, R, JavaScript, React, PHP, SQL, Rust.
- **Technical Skills:** Git, OOP, Data Structures, Algorithms, Linux, Tableau, Office 365 Products.